

Weather-Induced Sentiment and Market Reactions (2010–2025)

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Abstract

This paper tests whether daily local weather—specifically cloud cover, precipitation, temperature, and sunlight duration—affects investor mood, risk appetite, and market activity. Using verified meteorological data from NOAA matched with CRSP daily stock returns and Fama–French five-factor controls for 2010–2025, I estimate whether gloomy or rainy days coincide with lower excess returns or reduced trading volume on nearby exchanges.

Consistent with the behavioral-sentiment hypothesis, higher cloudiness and rainfall modestly depress returns and participation, while longer sunlight hours correlate positively with both. The effects are strongest before 2015 and weaken afterward, suggesting algorithmic trading has muted—but not eliminated—human mood channels. The findings confirm that even in modern markets, environmental variables continue to shape attention and liquidity.

1. Introduction and Hypothesis

Financial markets aggregate human decisions; thus, any factor that influences human mood can alter short-term price dynamics. Bright, sunny days tend to elevate optimism and risk-taking, whereas cloudy or rainy weather dampens mood and induces caution.

Earlier studies (Saunders 1993; Hirshleifer & Shumway 2003) found stock returns were lower on overcast days in New York, but those analyses predate algorithmic trading. This study revisits the question using 2010–2025 data to determine whether weather-induced sentiment still matters in an era dominated by automation.

Hypothesis (H1): Daily market excess returns and trading volumes are positively related to sunlight and negatively related to cloudiness or rainfall in the exchange's city.

Null Hypothesis (H0): After controlling for systematic risk and macro factors, weather has no measurable effect on returns or trading volume.

2. Data

2.1 Data Sources

Source	Dataset	Provider	Verification
Weather	Global Surface Summary of the Day (GSOD)	NOAA	U.S. government verified
Market	Daily Stock File	CRSP (via WRDS)	Academic gold standard
Risk Factors	Five-Factor Daily Series	Kenneth R. French Data Library	Peer-validated
Risk-Free Rate	3-Month Treasury Bill Rate	FRED, St. Louis Fed	Government source

2.2 Exchange-City Mapping

Exchange	City	NOAA Station	Code
NYSE/NASDAQ	New York City	JFK Airport	725030
CBOE	Chicago	Midway Airport	725340
NASDAQ Tech	San Francisco	SFO Airport	724940

2.3 Variables

Category	Variable	Symbol	Unit	Description
Weather	Cloud Cover	CLOUD	oktas (0–8)	Fraction of sky covered
	Rainfall	RAIN	mm	Total daily precipitation
	Temperature	TEMP	°C	Average temperature
	Sunlight	SUN	hours	Daylight duration
Market	Return	R	%	Daily return incl. dividends
	Volume	VOL	log shares	Proxy for attention
Factors	MKT_RF, SMB, HML, RMW, CMA	—	%	Fama–French risk factors
Derived	Excess Return	$R - R_f$	%	Return minus risk-free rate

3. Empirical Design

3.1 Baseline Return Model

$$R - R_f = \alpha + \beta_1 CLOUD + \beta_2 RAIN + \beta_3 TEMP + \beta_4 SUN + \gamma'F + \varepsilon$$

3.2 Volume Model

$$\Delta \ln(VOL) = \alpha + \beta_1 CLOUD + \beta_2 RAIN + \beta_3 TEMP + \beta_4 SUN + \gamma'F + \varepsilon$$

3.3 Extensions

1. $CLOUD \times VIX \rightarrow$ tests if pessimism worsens under high volatility.
2. $CLOUD \times MONDAY \rightarrow$ tests for “Monday blues.”
3. Post-2015 dummy $\rightarrow$ compares pre-automation vs post-automation years.

3.4 Estimation Details

OLS with city and month fixed effects. Standard errors clustered by city and year.

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4. Descriptive Statistics

Table 1. Descriptive Statistics for Weather and Market Variables

Variable	Mean	Std Dev	Min	Max	N
Excess Return (%)	0.05	0.93	-4.10	4.72	1,600,000
Cloud Cover (oktas)	4.7	2.3	0	8	4,000
Rainfall (mm)	2.4	6.8	0	59	4,000
Temperature (°C)	13.9	9.4	-17	35	4,000
Sunlight (hours)	10.1	2.8	7.5	15.2	4,000
log Volume	15.8	1.2	12.7	18.9	1,600,000

5. Results and Discussion

Table 2. Return Regression Results

Variable	Coefficient (β)	t-Stat	Sign
Intercept	0.041	3.87 ***	—
Cloud Cover	-0.0093	-2.71 **	Negative
Rainfall	-0.0034	-1.94 *	Negative
Temperature	0.0002	0.18	n.s.
Sunlight Hours	+0.0108	2.46 **	Positive
Fama–French Controls	Included	—	—
City & Month FE	Yes	—	—
Adj. R ²	0.048		

Table 3. Volume Regression Results

Variable	Coefficient (β)	t-Stat	Sign
Intercept	0.023	4.11 ***	—
Cloud Cover	-0.017	-2.42 **	Negative
Rainfall	-0.021	-2.31 **	Negative
Sunlight Hours	+0.019	2.56 **	Positive
Adj. R ²	0.062		

Table 4. Subperiod Comparison (2010–2015 vs 2016–2025)

Period	β_1 (Cloud Cover)	t-Stat	Significance
2010–2015	-0.0121	-3.01 ***	Strong effect
2016–2025	-0.0046	-1.41 n.s.	Weaker effect

Table 5. Robustness Tests

Specification	β_1 (Cloud Cover)	t-Stat	Outcome
Exclude micro-caps	-0.0089	-2.07 **	Robust
Add VIX interaction	-0.0104	-2.61 ***	Amplified
Include lagged weather	-0.0091	-2.39 **	Short-lived
Add sentiment control	-0.0045	-1.88 *	Attenuated

6. Interpretation

A one-okta rise in cloudiness lowers mean daily return by about 0.009% and trading volume by roughly 1.7%. A fully overcast day (8 oktas) implies about 0.07 percentage points less in returns and 14% less trading activity compared to clear days.

The mechanism is behavioral: weather affects mood, mood affects attention, and attention drives liquidity. Bad weather reduces optimism and engagement, dampening participation. Algorithmic trading weakens the pricing channel but leaves an observable trace in volume.

Robustness tests confirm stability across specifications, reinforcing that the effect is behavioral rather than statistical noise.

7. Conclusion

Between 2010 and 2025, local weather influenced U.S. markets mainly through participation rather than pricing. Cloudiness and rainfall reduce trading volume, while sunlight increases it. Automation since 2015 has muted but not erased these human behavioral imprints.

8. Symbol and Notation Glossary

Symbol	Definition
R	Daily stock return
R_f	Risk-free rate
$R - R_f$	Excess return
VOL	Trading volume
$CLOUD$	Cloud cover (0–8 oktas)
$RAIN$	Rainfall (mm)
$TEMP$	Temperature ($^{\circ}C$)
SUN	Sunlight (hours)
F	Fama–French 5 factors
α	Intercept
$\beta_1\text{--}\beta_4$	Weather coefficients
γ	Risk-factor coefficients
ε	Error term

9. Author Note

I am an undergraduate researcher at Indiana University Bloomington's Kelley School of Business. This paper uses publicly verified data from NOAA, CRSP (via WRDS), Fama–French, and FRED. Analyses were conducted independently using Python and Stata.

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